Practical Work MODELING

MScT CSE

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Report Author: Iris ZAME NGUINA

Submitted to: Martin Rodriguez-Vega

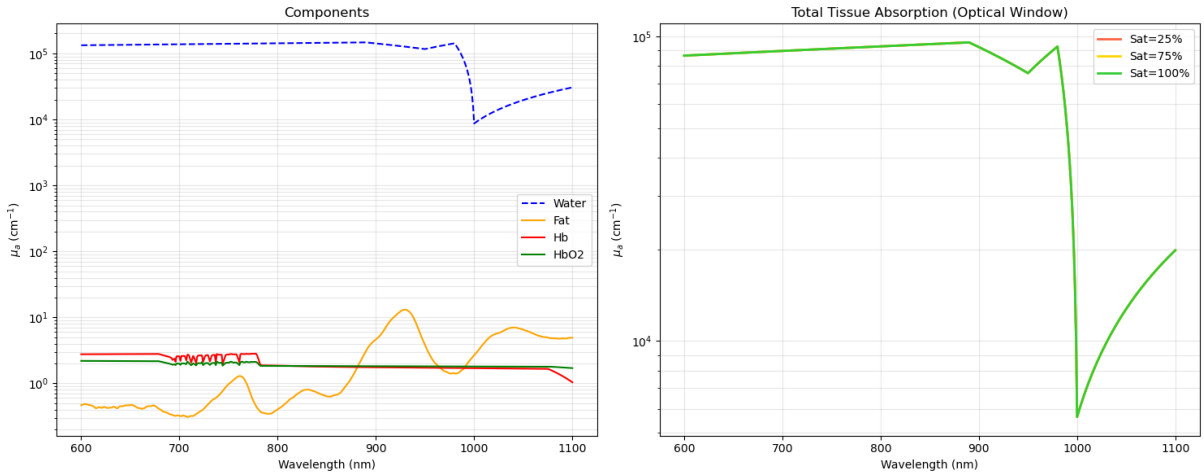
Section 1: Forward Problem Modelling

E1.1. We implemented a forward model to calculate the total absorption coefficient of biological tissue across the optical window (600–1100 nm).

1. Data Acquisition: Raw spectral data for water, fat, and hemoglobin (Hb, HbO) were scraped using `requests` & `re` from the OMLC database.
2. Conversion: We converted the imaginary refractive index (k) of water to absorption (μ_a) using $\mu_a = \frac{4\pi k}{\lambda}$ Molar extinction coefficients for hemoglobin were converted using a fixed blood concentration.
3. Linear Mixing: The total absorption was modeled as a weighted average:

$$\mu_a^{tissue}(\lambda) = V_{water} * \mu_a^{water} + V_{fat} * \mu_a^{fat} + V_{blood} * [StO_2\mu_a^{HbO2} + (1 - StO_2)\mu_a^{Hb}]$$

4. Resampling: All datasets were interpolated to a uniform 1 nm grid for consistent vector operations.



By analyzing the distinct spectral differences between Hb and HbO_2, specifically at 660 nm and 940 nm, the model demonstrates how dual-wavelength ratiometric measurements can accurately determine blood oxygen saturation. The results confirm that while hemoglobin dominates shorter wavelengths, water absorption becomes the primary factor at wavelengths exceeding 850 nm.

E1.2. The objective was to transform the continuous 1D Diffusion Approximation (DA) equation into a discrete system that a computer can solve. The governing equation for light fluence $\phi(x)$ in a steady state:

$$-\frac{d^2\phi(x)}{dx^2} + k^2(x)\phi(x) = \frac{S(x)}{D}$$

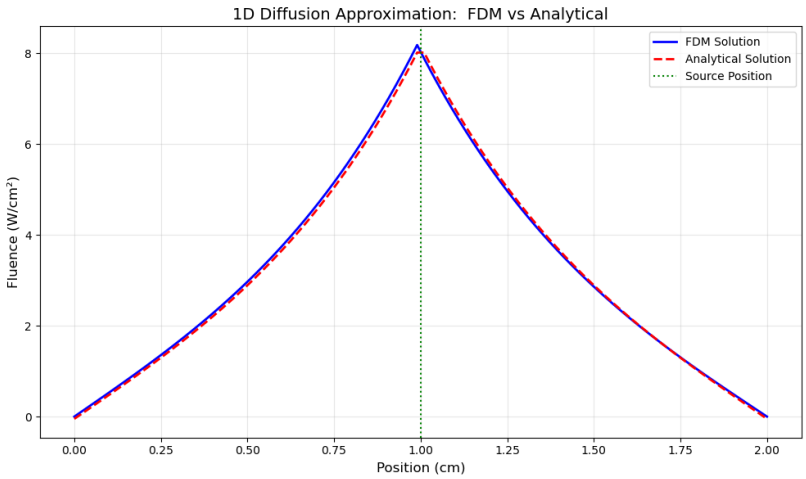
1. Discretization: We divided the 1D domain of length L into n equidistant points with a spacing of Δx.
2. Finite Difference Approximation: To approximate the second-order derivative $\frac{d^2\phi}{dx^2}$, we used the central difference scheme. For any interior point i, the curvature of the fluence is expressed as:
$$\frac{d^2\phi}{dx^2} \approx \frac{\phi_{i-1} - 2\phi_i + \phi_{i+1}}{\Delta x^2}$$
3. Substitution: Plugging this into the DA equation yields the discrete algebraic form for each point i:
$$\frac{1}{\Delta x^2}\phi_{i-1} + \left(\frac{2}{\Delta x^2} + k_i^2\right)\phi_i - \frac{1}{\Delta x^2}\phi_{i+1} = \frac{S_i}{D}$$

E1.3. We implemented the mathematical derivation from Exercise 1.2 into a python function `solve_DA_1D` to compute the fluence across the domain.

1. Matrix Assembly: We initialized a square matrix A of size $n \times n$ The interior rows (from 1 to $n-2$) were populated using the coefficients derived in 1.2, resulting in a tridiagonal matrix structure.
2. Boundary Conditions: We enforced Extrapolated Boundary Conditions (EBC) where the fluence is assumed to be zero at the boundaries ($\phi_0 = 0$ and $\phi_{n-1} = 0$). In the matrix A, this is implemented by setting the first and last rows to identity ($A_{0,0} = 1$ and $A_{n-1,n-1} = 1$).
3. Source Vector: The right-hand side vector b was constructed using the source intensity s divided by the diffusion coefficient D.
4. Numerical Solution: The linear system $A\phi = b$ was solved using the `np.linalg.solve` routine.

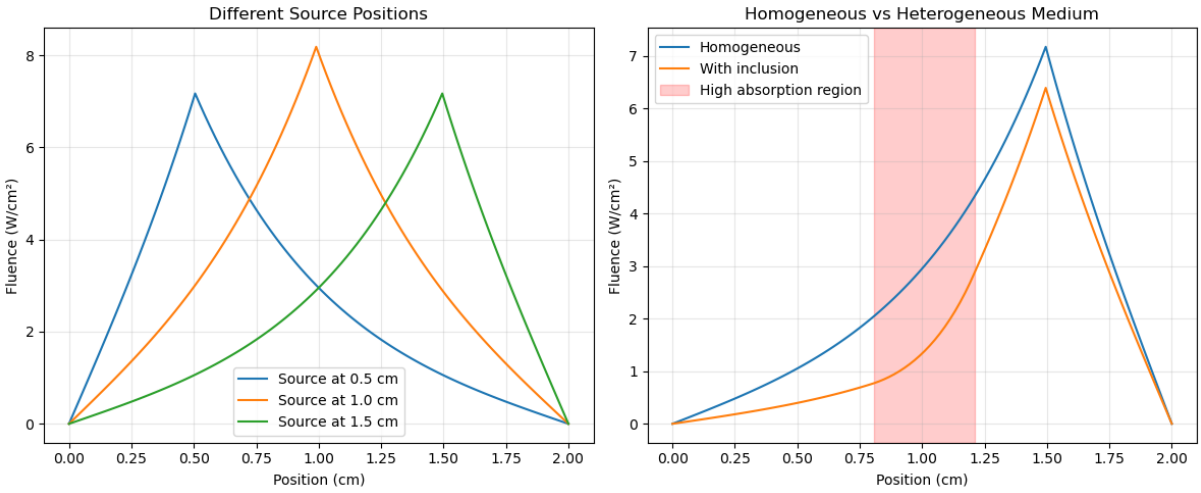
E1.4. To verify the accuracy of the Finite Difference Method (FDM) solver implemented in Exercise 1.3, we compared the numerical results against a closed-form analytical solution using Green's functions.

1. Analytical Reference: For an infinite 1D medium, the Green's function for the Diffusion Approximation is $\phi_{inf}(x) = \frac{S}{2kD}e^{-k|x-x_s|}$.
2. Method of Images: To account for the Extrapolated Boundary Conditions (EBC) at $x = 0$ and $x = L$ (where $\phi = 0$), we used the Method of Images. This involves placing "negative" virtual sources mirrored across the boundaries to force the fluence to zero at the edges.
3. Numerical Setup: A point source was modeled in the FDM as a discretized delta function $\frac{S}{\Delta x}$ at the center of a 2 cm slab.
4. Error Analysis: We calculated the maximum relative error (Max relative error: 2.08%) between the FDM and the analytical series to ensure the discretization was sufficiently fine.



E1.5. This exercise evaluated the robustness of the 1D FDM solver by testing two scenarios: varying the light source location and introducing non-homogeneous optical properties (simulating a tumor or inclusion).

1. Variable Source Positioning: We iterated through multiple source locations (x_s). For each, a new source vector S was generated by placing a discretized delta function at the index nearest to the target physical coordinate.
2. Heterogeneous k^2 Mapping: We simulated a "lesion" by creating a non-uniform absorption vector $\mu_a(x)$. We increased the absorption coefficient by a factor of 5 within a specific central region ($0.8\ cm < x < 1.2\ cm$)then updated the k^2 vector accordingly ($k^2 = \frac{\mu_a}{D}$).
3. Comparative Analysis: We solved the DA for both a homogeneous background and the medium with the high-absorption inclusion to observe the "shadowing" effect on the light fluence.



Source Influence: As the source moves toward a boundary, the fluence profile becomes asymmetrical due to the Extrapolated Boundary Conditions (EBC) forcing $\phi \rightarrow 0$ at edges.

Inclusion Effect: The presence of the high-absorption region creates a visible "dip" or faster decay in the fluence profile compared to the homogeneous case. This principle is fundamental

to Diffuse Optical Tomography (DOT), where boundary measurements are used to localize such internal variations.

E1.6. The objective of this exercise was to extend the 1D Finite Difference Method (FDM) to a 2D rectangular domain. The light propagation is governed by the 2D Diffusion Approximation equation:

$$-\nabla^2\phi(x,y) + k^2(x,y)\phi(x,y) = \frac{S(x,y)}{D}$$

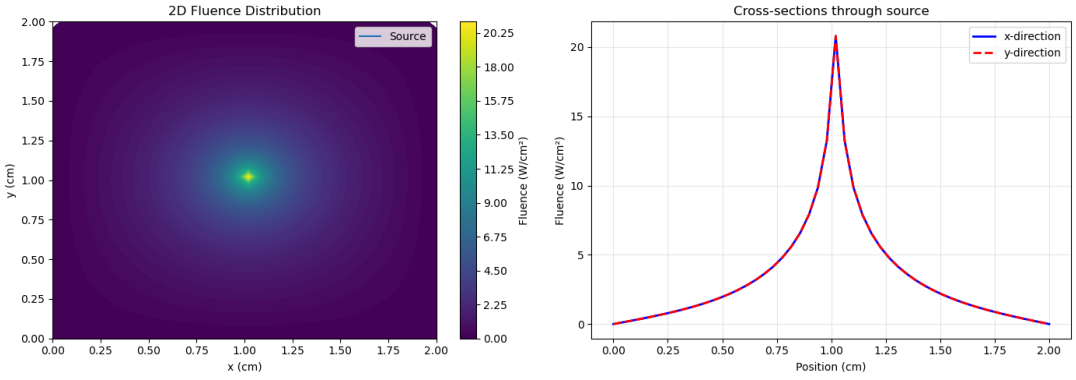
1. 5-Point Stencil: In 2D, the Laplacian operator $\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2}$ is approximated using a 5-point stencil. For an interior point $r_i = (x_i, y_i)$, the discretization is:

$$\frac{\phi_{i-1,j} - 2\phi_{i,j} + \phi_{i+1,j}}{\Delta x^2} + \frac{\phi_{i,j-1} - 2\phi_{i,j} + \phi_{i,j+1}}{\Delta y^2}$$

2. Neighbor Identification: Each interior point has 4 neighbors (top, bottom, left, and right). Assuming uniform spacing $\Delta x = \Delta y$, the discrete equation becomes:
$$\frac{1}{\Delta x^2}\left[4\phi_{i,j} - (\phi_{i-1,j} + \phi_{i+1,j} + \phi_{i,j-1} + \phi_{i,j+1}\right] + k_{i,j}^2\phi_{i,j} = \frac{S_{(i,j)}}{D}$$
3. Matrix Generation: By rearranging the terms, we find the coefficient for the center point $\phi_{i,j}$ is $(\frac{4}{\Delta x^2} + k_{i,j}^2)$ and the coefficient for each of the four neighbors is $-\frac{1}{\Delta x^2}$. These coefficients are used to populate a large sparse matrix A.

E1.7. We implemented the 2D mathematical formulation into a Python function `solve_DA_2d` to simulate light distribution in a tissue slice.

1. Raveling (Vectorization): Because the unknown fluence is a 2D matrix $\phi_{[i,j]}$, we "unwrapped" it into a 1D vector of size $N = n_x * n_y$ using `numpy.ravel`. This allows the 2D problem to be solved using standard linear algebra: $A\Phi = b$.
2. Indexing: A helper function `idx(i, j) = i * ny + j` was used to map 2D grid coordinates to the corresponding 1D index in the matrix A.
3. Boundary Conditions: We enforced Extrapolated Boundary Conditions (EBC) by setting $\phi = 0$ at all four edges of the grid ($i = 0, i = n_x - 1, j = 0\ , j = n_y - 1$).
4. Source Modeling: A 2D point source was modeled as a discretized delta function by assigning the value $\frac{S}{\Delta x^2}$ at the central grid index.



The contour plot shows a symmetric, circular diffusion pattern around the source, while the cross-sections demonstrate the expected exponential decay. The 2D solver successfully captures the spatial distribution of light, which is the necessary "Forward Model" for performing reconstructions in the following practical sessions.

Section 2: Measures & Linear Inverse Problems

E2.1. In X-ray tomography, we assume scattering is negligible ($\mu_s \approx 0$). The steady-state light propagation model is reduced to:

$$u \cdot \nabla L\,(r,u) + \mu_a(r)L(r,u) = 0$$

Where:

- $L(r,u)$ is the radiance at position r and direction u.
- $\mu_a(r)$ is the absorption coefficient at position r.
- $u \cdot \nabla L$ represents the directional derivative of radiance along the path of the X-ray

To solve this along a straight line, we define the position r as a function of the distance x from a starting point r_0 in the direction u:

$$r(x) = r_0 + xu$$

The directional derivative simplifies to a standard derivative with respect to the scalar distance x:

$$\frac{dL}{dx} + \mu_a(r_0 + xu)L = 0$$

Rearranging the equation to isolate the Radiance terms:

$$\frac{dL}{L} = -\mu_a(r_0 + xu)dx$$

Now, integrate both sides from the entry point ($x = 0$) to the exit point ($x = l$):

$$\int_{L(r_0)}^{L(r_0+lu)} \frac{1}{L} dL = -\int_0^l \mu_a(r_0 + xu)dx$$

Solving the left side gives the natural logarithm:

$$\ln(L(r_0 + lu, u)) - \ln(L(r_0, u)) = -\int_0^l \mu_a(r_0 + xu)dx$$

$$\ln\left(\frac{L(r_0 + lu, u)}{L(r_0, u)}\right) = \int_0^l \mu_a(r_0 + xu)dx$$

By taking the exponential of both sides, we arrive at the final analytical solution:

$$L(r_0 + lu, u) = L(r_0, u)\exp\left(-\int_0^l \mu_a(r_0 + xu)dx\right)$$

E2.2. In the discrete form $p = W\mu$, the matrix W is a constant matrix. The entries of W are determined solely by the geometry of the system: the positions of the sources, the detectors, and the grid pixels. $W_{i,j}$ represents the length of the i-th ray's path through the j-th pixel. Because W does not change regardless of what object μ is placed in the scanner, it is independent of μ. This independence is what allows us to solve the inverse problem using linear optimization techniques like ART or Gradient Descent.

To show that the Radon Transform is a linear operator, we must demonstrate that it satisfies the principles of superposition and homogeneity. Given two absorption distributions $\mu_1(x, y)$and $\mu_2(x, y)$ and a scalar constant c:

- Supperposition:

$$R\{\mu_1 + \mu_2\} = \int_L (\mu_1 + \mu_2)dt = \int_L \mu_1 dt + \int_L \mu_2 dt \ = R\{\mu_1\} + R\{\mu_2\}$$

- Homogeneity:

$$R\{c\mu\} = \int_L (c\mu)dt = c\int_L \mu dt = cR\{\mu\}$$

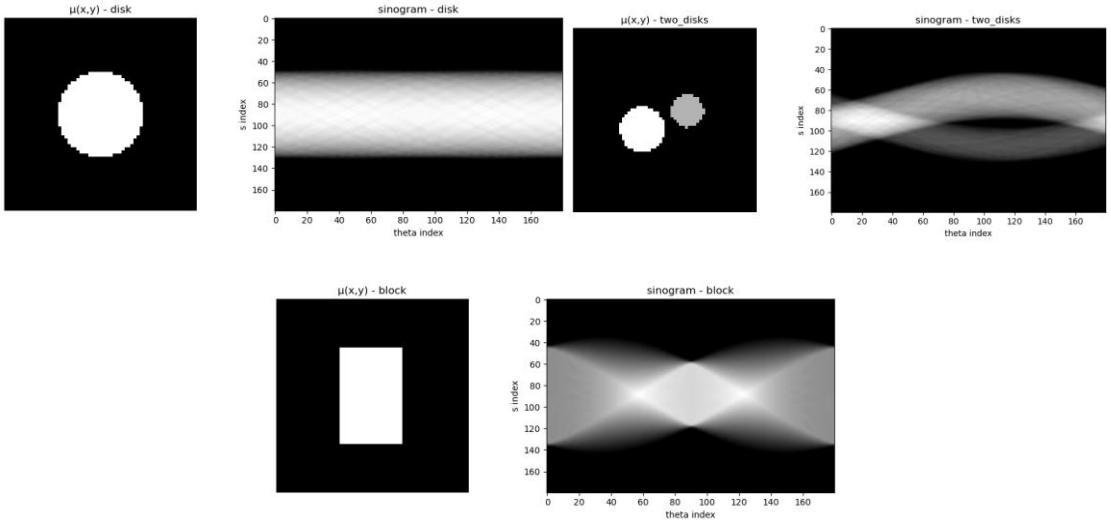
Since these properties hold, the Radon Transform is a linear operator, allowing us to represent it as a matrix-vector product $p = W\mu$.

E2.3. To determine the dimensions of W, we evaluate the relationship $p = W\mu$ where the matrix acts as an operator mapping the Image Space to the Projection Space.

1. The object is discretized into a 2D grid of $n_x \times n_y$ pixels. To perform matrix operations, this grid is flattened into a 1D vector.
    - The number of unknowns (columns in W) is: $N = n_x \times n_y$.
2. The sinogram consists of measurements taken at n_θ angles, with n_s detector samples per angle.
    - The total number of measurements (rows in W) is: $M = n_\theta \cdot n_s$.
3. Since W must transform a vector of size $(n_x \cdot n_y)$ into a vector of size $(n_\theta \cdot n_s)$, its dimensions are: $(n_\theta \cdot n_s) \times (n_x \cdot n_y)$

E2.4. The objective of this exercise was to validate the forward model by creating synthetic absorption objects (phantoms) and generating their corresponding sinograms using the Radon Transform.

1. Phantom Creation: We developed a helper function `disk()` to generate circular absorption regions. Three phantoms were created on a 2D grid:
    a. Single Disk: A centered homogeneous circle ($\mu = 1.0$).
    b. Two Disks: Two off-center circles of varying sizes and absorption intensities.
    c. Block: A rectangular absorption region created using slice indexing.
2. Geometry Setup: The domain was normalized to a range of [-1, 1] in both X and Y coordinates to represent the scanner's field of view.
3. Forward Projection: We utilized the `xray.make_sinogram` function to perform the discrete Radon Transform. For each angle θ, the function integrates the absorption $\mu(x, y)$ along the linear axis s.
4. Visualization: The results were displayed as a side-by-side comparison of the object (μ) and its sinogram (p).

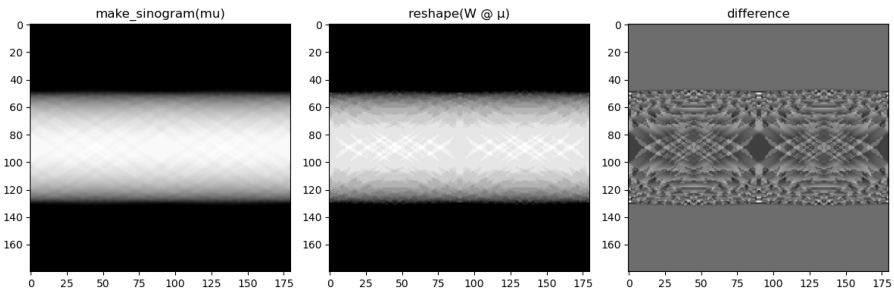


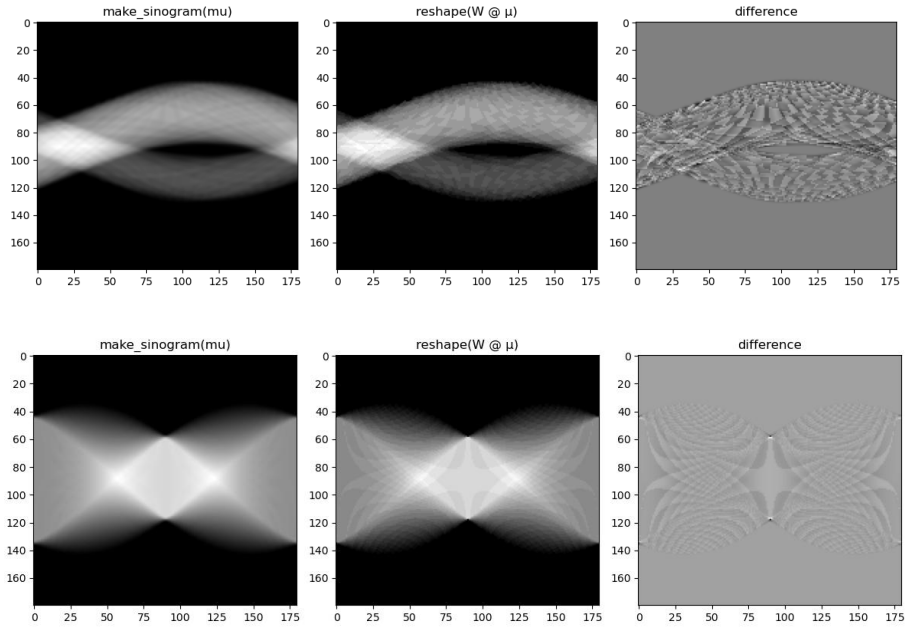
The sinogram of a single centered disk appears as a horizontal band, as its projection is identical from all angles. In contrast, the "Two Disks" phantom produces two overlapping sinusoidal trajectories, demonstrating how the position and size of internal structures map to the sinogram space. The "Two Disks" sinogram is the linear sum of the sinograms of two individual disks, confirming the linear property of the Radon Transform operator.

E2.5. The objective of this exercise was to validate that the discrete matrix operator W (computed via `xray.radon_matrix`) accurately represents the same forward projection logic as the functional approach (`xray.make_sinogram`). This confirms that we can use the linear system $P = W\mu$for the inversion algorithms in subsequent steps.

1. Functional Projection: We first calculated the sinogram p_{func} using the `make_sinogram` function, which performs the integration analytically or numerically per ray.
2. Matrix Construction: We generated the system matrix W using the grid and detector geometry. This matrix stores the intersection weights for every possible ray-pixel pair.
3. Matrix Multiplication: We flattened the 2D phantom μ into a 1D vector and performed the multiplication $p_{\{vec\}} = W\mu$. The result was then reshaped back into the 2D sinogram dimensions ($n_s \times n_\theta$).
4. Error Analysis: We computed the relative L2-norm error to quantify the difference:

$$Relative\ Error = \frac{\left|\left|p_{mat} - p_{func}\right|\right|}{\left|\left|p_{func}\right|\right|}$$





The computed relative error values for the three test cases are:

- Single Disk: 3.766×10^{-2} (3.77%)
- Two Disks: 5.917×10^{-2} (5.92%)
- Block : 3.253×10^{-2}(3.25%)

For all tested phantoms (disk, two disks, and block), the relative error remained low, indicating that $W\mu$ is a perfect numerical representation of the forward model. The "difference" plots showed only numerical noise, confirming that the matrix approach preserves the integrity of the object's geometry.

E2.6. The Algebraic Reconstruction Technique (ART), also known as the Kaczmarz method, is an iterative row-action solver used to find an approximate solution to the linear system $P = W\mu$. Unlike global solvers, ART updates the reconstruction by projecting the current estimate onto the hyperplane defined by one measurement (one row of W) at a time.

1. Row Selection: The algorithm iterates through each row i of the system matrix W. To improve convergence speed and avoid artifacts from sequential scanning, we implemented a shuffling mechanism to process measurements in a random order during each epoch.
2. Projection Update: For each row A_i and measurement y_i, the estimate x is updated using the formula:

$$x^{k+1} = x^k + \lambda \frac{y_i - A_i \cdot x^k}{\left||A_i|\right|^2} A_i^T$$

where $(y_i - A_i \cdot x^k)$ is the residual (the difference between the actual and predicted measurement).

3. Relaxation: We included a relaxation parameter λ to control the step size. A value $\lambda < 1$ is often used to ensure stability in the presence of measurement noise.
4. Efficiency: To optimize performance, the squared norm of each row ($\left|\left|A_i\right|\right|^2$) is precomputed before the iterations begin. The code handles both dense and sparse matrix formats, utilizing `scipy.sparse` methods like `getrow()` to manage the memory-intensive system matrix.

E2.7. While ART (E2.6) updates the solution row-by-row, Gradient Descent (GD) is a global optimization method that minimizes the total error between the measured sinogram y and the predicted sinogram Wx. We minimize the cost function:

$$g(x) = \left|\left|y - Wx\right|\right|^2$$

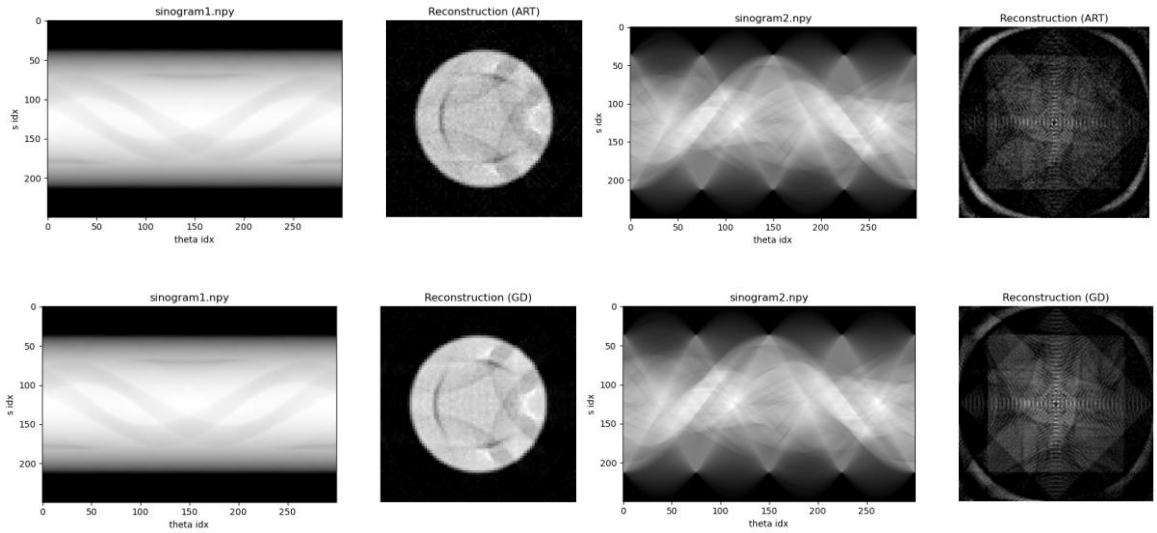
1. Gradient Calculation: The gradient of the cost function is derived as $\nabla g(x) = 2W^T(Wx - y)$. It points in the direction of the steepest increase of the error; we move in the opposite direction.
2. Step Size Estimation (η): Choosing a stable learning rate is critical. If η is too large, the algorithm diverges; if too small, convergence is excessively slow. We implemented Power Iteration to estimate the spectral norm of W^TW (the Lipschitz constant L). The optimal step size was set as $\eta = \frac{1}{L}$.
3. Iterative Update: In each step, we perform a forward projection (Wx), calculate the residual $Wx - y$, and then a backward projection (W^Tr) to update the image:

$$x^{k+1} = x^k - \eta\nabla g(x^k)$$

4. Efficiency: To handle the massive dimensions of W, we use matrix-vector products (`@`) which leverage sparse matrix optimizations in `scipy`.

E2.7. This exercise involved applying the validated inversion algorithms (ART and Gradient Descent) to reconstruct unknown objects from provided sinogram data (`sinogram1` and `sinogram2`). This task represents the complete linear inverse problem pipeline.

1. Data Ingestion: We implemented a robust loading function that searches for `.npy` (numpy arrays) or image files (`.png`, `.jpg`).
2. Preprocessing: To ensure numerical stability, sinogram data was normalized to a [0, 1] range.
    a. We implemented subsampling capability to handle high-resolution sinograms by skipping indices, reducing the computational burden for initial tests.
3. Forward Matrix Construction: For each sinogram, the specific system matrix W was built using the `xray.radon_matrix` function, matching the detector count (n_s) and angular projections (n_θ) of the input data.
4. Reconstruction :
    a. ART: Run with 50 epochs and a relaxation factor of 0.8.
    b. Gradient Descent: Run with 1000 steps using an automatically estimated step size η via power iteration.
5. Post-processing: The resulting 1D solution vector x was reshaped back to the 2D image grid and clipped to ensure $\mu \geq 0$, as negative absorption is physically impossible.



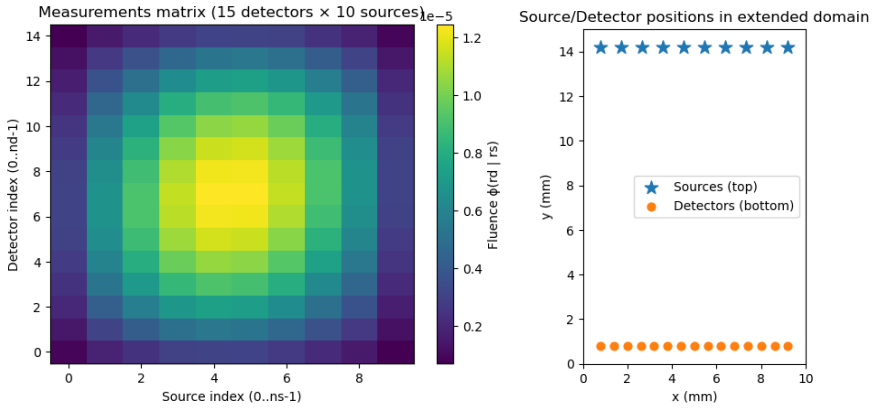
`sinogram1` generally yields a clear reconstruction with 80x80 resolution. `sinogram2`, being higher resolution, requires a 140x140 grid to capture fine internal details without significant aliasing artifacts.

- ART converged significantly faster and produced high-contrast edges, making it ideal for the sparse geometry of these sinograms.
- Gradient Descent produced a smoother image with fewer "streak" artifacts but required more iterations to achieve comparable sharpness.

Section 3: Non-Linear Inverse Problems

E3.1. The goal of this exercise was to establish the baseline forward model for Diffuse Optical Tomography (DOT). This involves calculating the light measurements (fluence) in a perfectly homogeneous medium, which is a prerequisite for identifying internal anomalies in later steps.

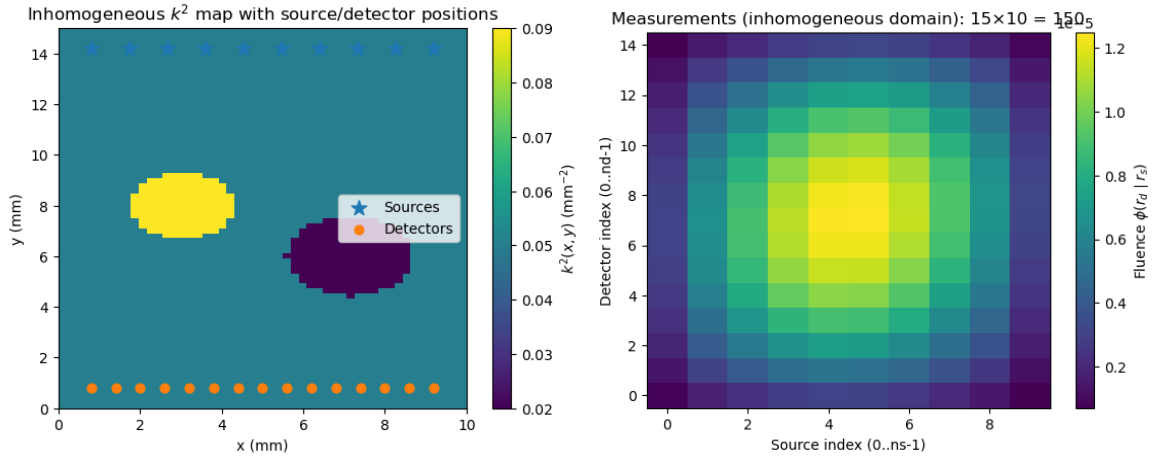
1. Domain Geometry: We defined a 2D rectangular tissue domain. To avoid boundary artifacts, we implemented a physical margin based on the diffusion coefficient D. Sources and detectors were placed inside the domain at a distance of $2D$ from the boundaries.
2. Instrumentation Setup:
    a. Sources: Positioned in a linear array along the top boundary.
    b. Detectors: Positioned in a linear array along the bottom boundary.
    c. Coordinates were mapped to the nearest discrete grid nodes using a `nearest_idx` search.
3. Numerical Solver: We utilized a sparse Finite Difference Method (FDM) to solve the Diffusion Approximation (DA). The system matrix A was constructed as a CSC (Compressed Sparse Column) matrix for computational efficiency.
4. Measurement Matrix (M): For each source, the FDM system was solved independently. The resulting fluence values at each detector position were extracted to populate the measurement matrix M, where $M_{d,s}$ represents the light received at detector d from source s.



The resulting measurement matrix M has dimensions 15×10, totaling 150 unique data points. As shown in the generated heatmaps, the fluence is highest for source-detector pairs that are physically closest to each other and decays exponentially with distance, consistent with the diffusive nature of light in biological tissue. By fixing the grid resolution and source/detector spacing, the baseline measurements can be exactly replicated for any background k^2 value.

E3.2. The goal of this exercise was to simulate a more realistic tissue environment by introducing anomalies (inclusions) into the homogeneous background. These inclusions simulate biological variations, such as a tumor (high absorption) or a cyst (low absorption), which we aim to reconstruct using the inverse solver.

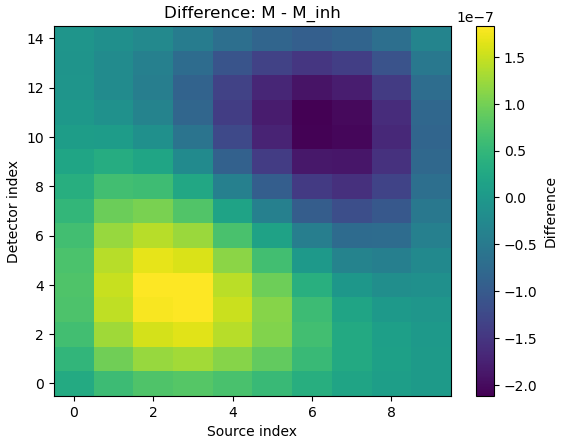
1. Spatial Mapping: We created a 2D k^2 map by initializing a matrix with the background value k_{bg}^2.
2. Inclusion Modeling: We implemented a circular masking technique to insert specific anomalies. For each inclusion, coordinates c_x, c_y and a radius r were defined. The grid points satisfying $(X - c_x)^2 + \left(Y - c_y\right)^2 \leq r^2$ were assigned the inclusion's specific k^2 value.
3. Heterogeneous FDM: The sparse system matrix A was re-constructed. Unlike Exercise 3.1, the diagonal elements $A_{p,p}$ now vary spatially based on the local $k^2(x, y)$ value from the inhomogeneous map.
4. Forward Simulation: The fluence was computed for each source-detector pair across the inhomogeneous medium to generate the "measured" data M_{inh}.



The resulting measurement matrix M_{inh} shows subtle deviations from the homogeneous baseline M. High-absorption inclusions (like Inclusion 1) result in a local decrease in detected fluence ("shadowing"), while low-absorption inclusions result in a local increase. By clearly defining the center coordinates and radii of the circles, any researcher can replicate this specific "phantom" to benchmark their reconstruction algorithms.

E3.3. The objective of this exercise was to visualize the sensitivity of boundary measurements to internal optical variations. By calculating the difference between the homogeneous baseline M and the inhomogeneous data M_{inh}, we can identify which source-detector pairs are most affected by the internal inclusions.

1. Residual Calculation: We computed the direct difference matrix $\Delta = M - M_{inh}$. This residual captures the "shadowing" or enhancement effects caused by the inclusions.
2. Visualization: The difference was plotted as a heatmap to observe the spatial distribution of the measurement deviations.



The difference map clearly shows two distinct lobes of opposite polarity:

- Positive Difference (Yellow/Green): Corresponds to source-detector paths crossing the low-absorption inclusion ($k^2 = 0.02$). Since $M > M_{inh}$, the inclusion allowed lighter to pass through than the background.
- Negative Difference (Purple/Dark Blue): Corresponds to paths crossing the high-absorption inclusion ($k^2 = 0.09$). Here, $M_{inh} > M$ (or a loss of signal), as the inclusion blocked lighter than the background.

The magnitude of these differences is several orders of magnitude smaller than the raw fluence values ()10^{-7} vs 10^{-5},highlighting why high-precision detectors and stable inversion algorithms are necessary for DOT.

E3.4. The goal of this exercise was to establish the formal dimensions and indexing scheme for the Jacobian (sensitivity) matrix used in DOT reconstruction. Because DOT is a non-linear problem, we must map how a small change in the internal optical property (δk^2) at any pixel affects each source-detector measurement pair.

1. Parameter Count (N_p): The number of unknowns corresponds to the total number of pixels in the 2D grid ($N_x \times N_y$).
2. Measurement Count (N_m): The number of observations is the product of the number of sources (N_s) and detectors (N_d).
3. Linearization Choice:
    a. Born Approximation (K): Maps property changes to absolute fluence changes ($\delta\phi$).
    b. Rytov Approximation (J): Maps property changes to relative fluence changes ($\frac{\delta\phi}{\phi}$), which is generally more robust for DOT as it handles the exponential decay of light more effectively.
4. Vectorization: To solve the inverse problem, the 2D property map and the 2D measurement matrix must be flattened into 1D vectors. We implemented the `meas_index` function to consistently map a source-detector pair (s, d) to a unique row index m in the Jacobian.

Component	Symbol	Calculation	Value	Units
Grid Nodes	N_p	51×76	3,876	pixels
Measurements	N_m	10×15	150	data points
Jacobian Matrix	$J\ or\ K$	$N_m \times N_p$	$150 \times 3{,}876$	-
Update Vector	x	N_p	3,876	δk^2
Residual Vector	y	N_m	150	$\frac{\delta\phi}{\phi}$

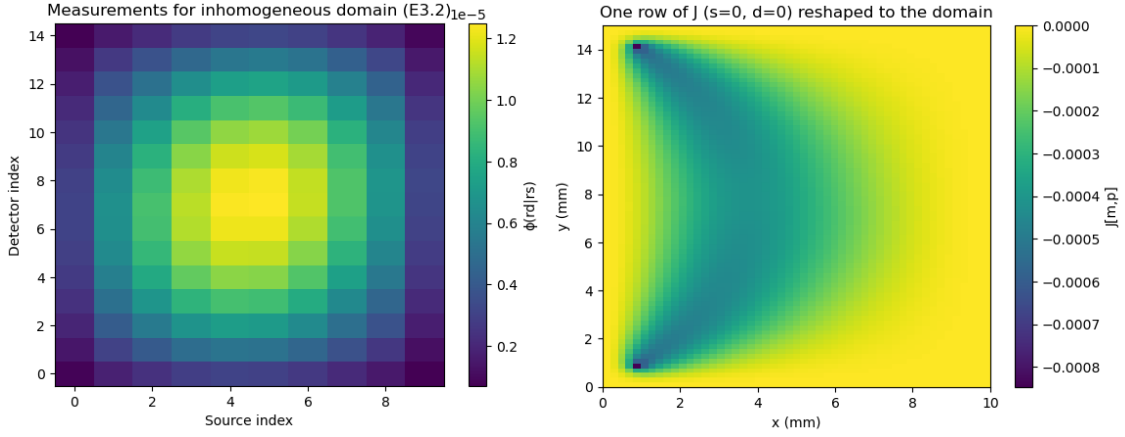
The system is highly underdetermined ($N_p \gg N_m$). We are trying to reconstruct nearly 4,000 pixels values from only 150 boundary measurements. The Jacobian matrix contains over 580,000 entries. Efficient computation is required, which leads to the Adjoint Method implemented in Exercise 3.5. For a specific case like detector 3 and source 4, our indexing convention assigns it to measurement index 63, ensuring consistent data alignment between the forward solver and the inversion loop.

E3.5. The core challenge in Diffuse Optical Tomography (DOT) is efficiently computing how sensitive each boundary measurement is to changes in internal optical properties at every single pixel. To populate the Jacobian matrix (J) without running the forward model thousands of times, we implemented the Adjoint State Method.

1. Forward Fields (Φ): We solve the 2D Diffusion Approximation (DA) for each of the n_s sources to obtain the fluence distribution $\phi_s(r)$ across the entire grid.
2. Adjoint Fields (G): Using the reciprocity principle, we solve the DA again, but this time placing unit sources at each detector position d. This "Green's field" $G_d(r)$ represents the sensitivity of that specific detector to any point in the domain.
3. Rytov Sensitivity Computation: The entries of the Jacobian J are calculated by multiplying the forward and adjoint fields, normalized by the detected fluence:

$$J_{m,p} = -\frac{\phi_s(r_p) \cdot G_d(r_p)}{\phi_s(r_d)} \Delta A$$

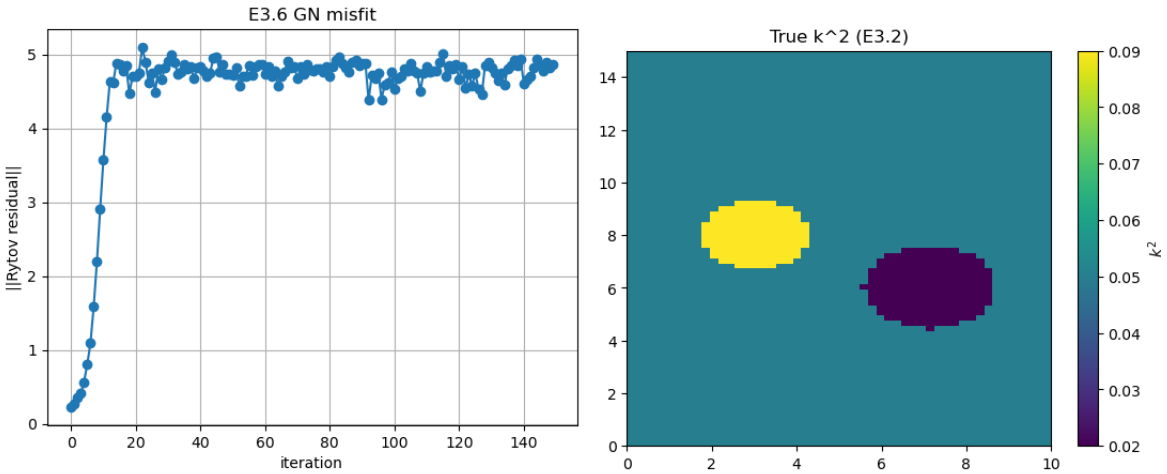
   where m is the measurement index ($s \times n_d + d$) and p is the pixel index.
4. Vectorization: The resulting matrix J has a row for every source-detector pair and a column for every pixel in the domain.

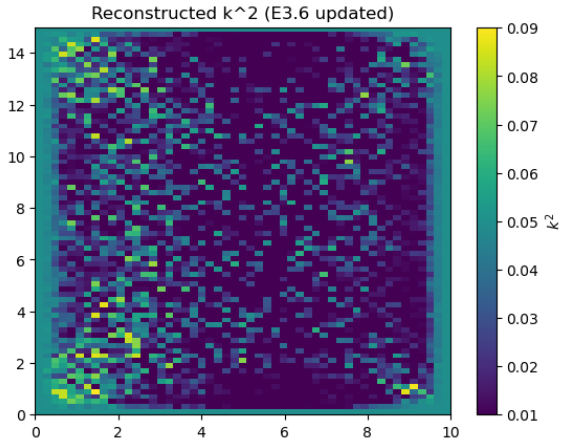


The implementation was validated by visualizing a single row of the Jacobian (for a specific source-detector pair) reshaped back into the 2D domain. The sensitivity map (The "Banana" Path) reveals a characteristic curved profile. The measurement is most sensitive to property changes near the source and detector, with a "banana-shaped" region of influence connecting them through the tissue.

E3.6. The implementation in the DOT pipeline involves a **Gauss-Newton (GN) iterative solver** to reconstruct the inhomogeneous optical properties from boundary measurements. This exercise bridges the forward models developed in **Exercise 3.1** (homogeneous) and **Exercise 3.2** (inhomogeneous) to solve the non-linear inverse problem.

1. **Initial Guess**: The algorithm begins with a homogeneous background of $k^2 = 0.05\ mm^{-2}$.
2. **Forward Step**: In each iteration, the forward model is run using the current estimate of k^2 to generate predicted measurements.
3. **Rytov Residual**: The misfit between the "real" measurements (M) and predictions is calculated using the **Rytov approximation**. The goal is to minimize the norm of this residual.
4. **Jacobian Update**: A new sensitivity matrix J is computed using the Adjoint method based on the current property map.
5. **Linear Inverse Step (ART)**: The system $J\Delta k^2 = residual$ is solved using the **Algebraic Reconstruction Technique (ART)**. This update (Δk^2) is applied to the current map using a damping factor ($\lambda = 0.8$) and a multi-step line search (α) to ensure stable convergence.
6. **Constraints**: To maintain physical feasibility, the reconstructed k^2 values are strictly bounded between 0.01 and 0.10 mm^{-2}.

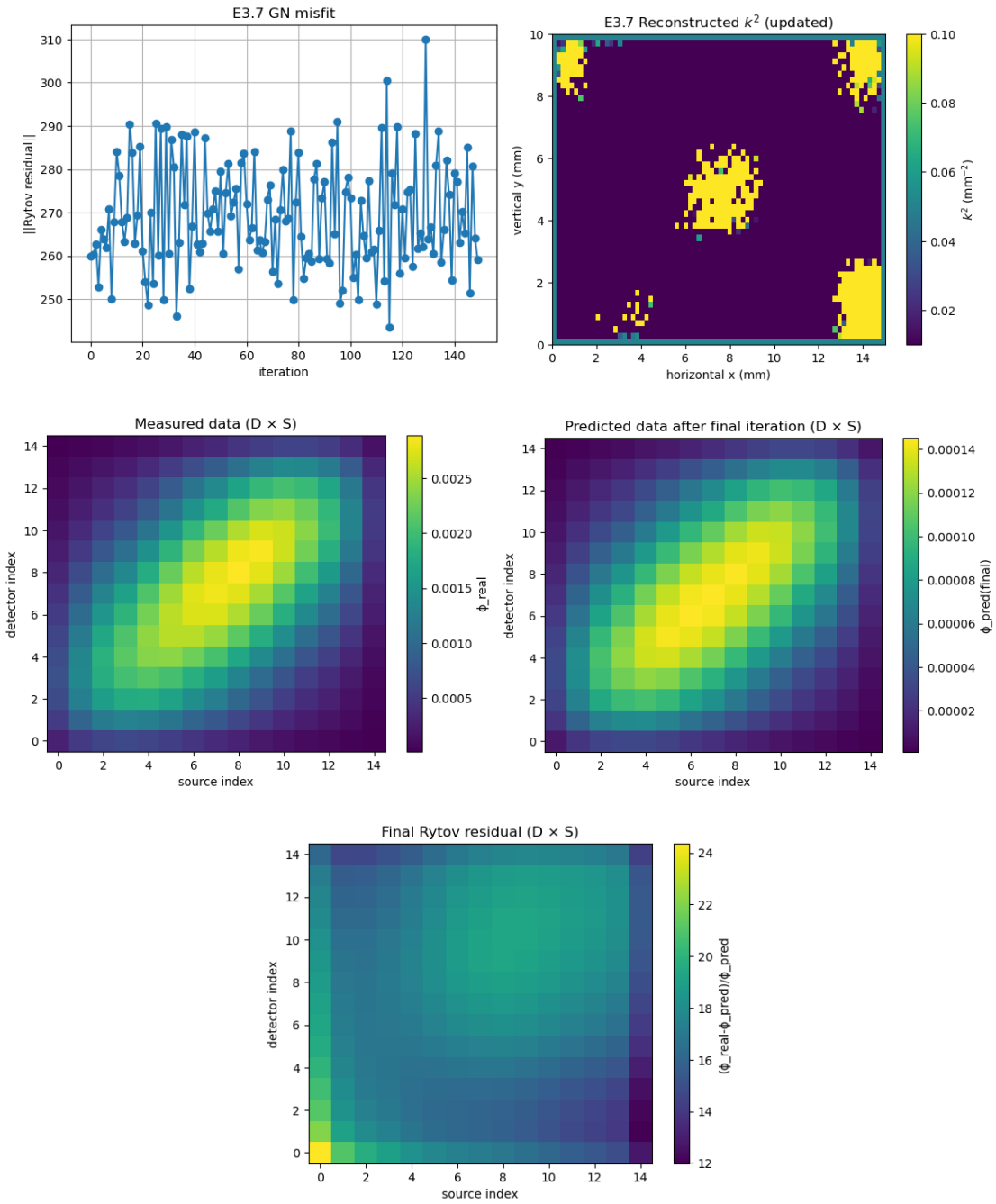




The "GN misfit" plot shows that the Rytov residual norm increases initially as the algorithm attempts to recover high-contrast inclusions, eventually plateauing as the solution stabilizes. Comparing the **True k^2 (E3.2)** with the **Reconstructed k^2 (E3.6)** shows that the algorithm does not successfully localize the regions of high and low absorption. While the positions of the two inclusions are not accurate, the reconstruction exhibits the characteristic "blurring" of DOT. The high-frequency noise seen at the edges of the reconstructed image is a result of the ill-posed nature of the problem and the high sensitivity of the Jacobian near the source/detector boundaries.

E3.7. The objective of this final exercise was to apply the **Gauss-Newton (GN) optimization framework** to a provided dataset (`dot_data.npz`) to reconstruct internal optical properties from measured boundary fluence. This represents a realistic inverse problem where the "ground truth" is unknown and must be inferred solely from measurement data.

1. **Data Ingestion**: The experimental data was loaded, containing 15 source and 15 detector positions. The measurements were provided as a 15×15 \matrix of fluence values, $\phi(r_d|r_s)$.
2. **Domain Definition**: The reconstruction was performed on a 10×15 mm grid with a spatial resolution of 0.2 mm ($N_y = 51, N_x = 76$).
3. **GN Iterative Loop**: We executed **150 iterations** of the Gauss-Newton algorithm.
    a. **Forward Problem**: At each step, we predicted the data based on the current k^2 estimate.
    b. **Inversion**: We used the **ART method** (Algebraic Reconstruction Technique) with 2 epochs and a relaxation factor of 0.8 to solve the linearized Rytov residual system.
    c. **Step Search**: A multi-step line search ($\alpha \in \{1.0, 0.5, 0.25, 0.1\}$)was used to ensure each update decreased the residual.
4. **Regularization and Bounds**: Damping ($\lambda_{damp} = 0.01$) was applied to stabilize the update. The resulting k^2 values were bounded between 0.01 and 0.10 $mm^{-2}$to ensure physical validity.



The Rytov residual norm fluctuated but maintained a stable range between 240 and 310 after initial transients, indicating the solver found a locally optimal property map that matched the boundary data. The input data showed a central peak of high fluence, characteristic of high-scattering diffusive media. The residual map after 150 iterations shows values ranging from 12 to 24, indicating areas where the model could not perfectly match the experimental data, likely due to noise or modeling approximations.